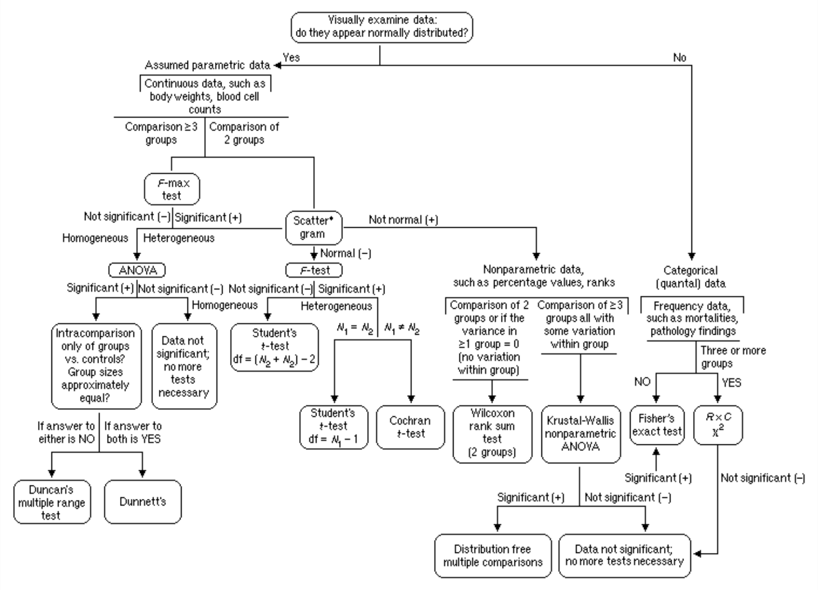
GLM as a unified framework for data analysis

Francisco Rodríguez-Sánchez

https://frodriguezsanchez.net

Statistics as a maze



So many questions

- **Why** should we really use analysis Y over Z?

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- What if they are **not independent**?

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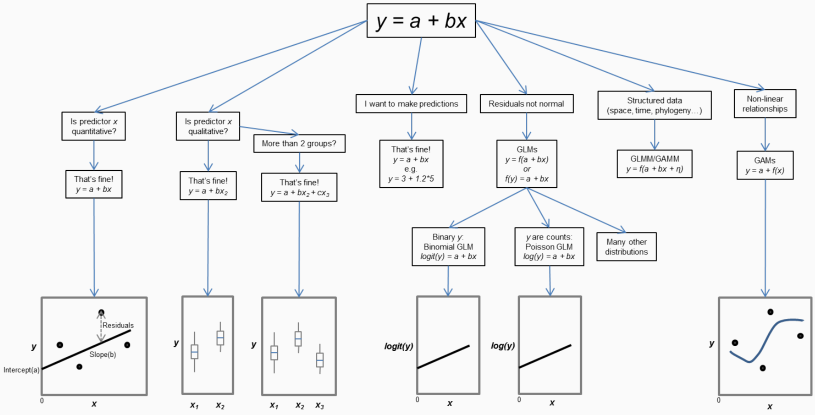
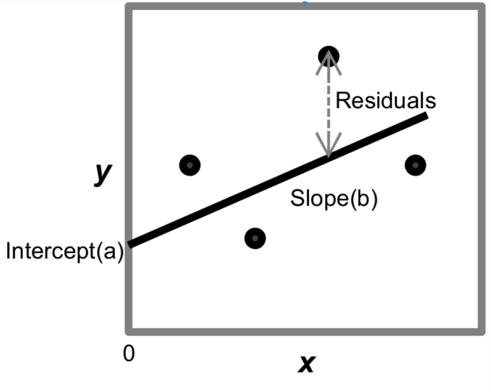
- **Why** should we really use analysis Y over Z?
- What if my data are **not Normal**?
- What if they are **not independent**?
- Why am I getting **different p-values** with different tests?
- What even is a **p-value**?
- How can I take **different factors** into account?
- Can I make **predictions**?

A unified framework



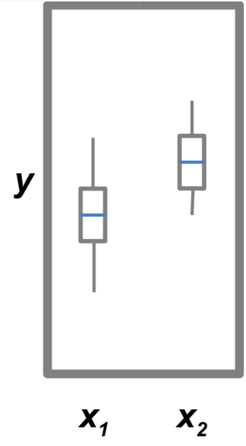
Linear regression

$$y = a + bx$$



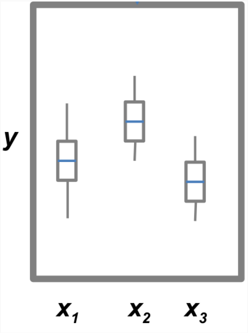
Is predictor X qualitative?

$$y = a + bx_2$$



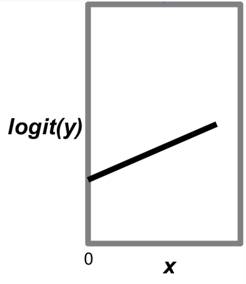
More than 2 groups?

$$y = a + bx_2 + cx_3$$



My data (residuals) are not Normal

$$y = f(a + bx)$$

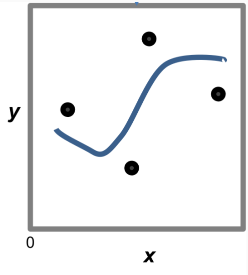


My data are structured (space, time, phylogeny)

$$y = f(a + bx + \eta)$$

Relationships are not linear

$$y = a + f(x)$$



t-tests

ANOVA

regression

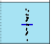
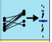
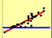
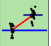
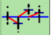
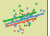
...

are special cases of GLM

Common statistical tests are linear models

Last updated: 02 April, 2019

See worked examples and more details at the accompanying notebook: https://lindeloev.github.io/tests-as-linear

	Common name	Built-in function in R	Equivalent linear model in R	Exact?	The linear model in words	Icon
Simple regression: lm(y ~ 1 + x)	**y is independent of x** P: One-sample t-test N: Wilcoxon signed-rank	t.test(y) wilcox.test(y)	lm(y ~ 1) lm(signed_rank(y) ~ 1)	✓ for N >14	One number (intercept, i.e., the mean) predicts **y**. - (Same, but it predicts the *signed rank* of **y**.)	
	P: Paired-sample t-test N: Wilcoxon matched pairs	t.test(y_1, y_2, paired=TRUE) wilcox.test(y_1, y_2, paired=TRUE)	lm(y_2 - y_1 ~ 1) lm(signed_rank(y_2 - y_1) ~ 1)	✓ for N >14	One intercept predicts the pairwise **y_2-y_1** differences. - (Same, but it predicts the *signed rank* of **y_2-y_1**.)	
	y ~ continuous x P: Pearson correlation N: Spearman correlation	cor.test(x, y, method='Pearson') cor.test(x, y, method='Spearman')	lm(y ~ 1 + x) lm(rank(y) ~ 1 + rank(x))	✓ for N >10	One intercept plus **x** multiplied by a number (slope) predicts **y**. - (Same, but with *ranked* **x** and **y**)	
	y ~ discrete x P: Two-sample t-test P: Welch's t-test N: Mann-Whitney U	t.test(y_1, y_2, var.equal=TRUE) t.test(y_1, y_2, var.equal=FALSE) wilcox.test(y_1, y_2)	lm(y ~ 1 + G_2)[A] gls(y ~ 1 + G_2, weights=…[B])[A] lm(signed_rank(y) ~ 1 + G_2)[A]	✓ ✓ for N >11	An intercept for **group 1** (plus a difference if **group 2**) predicts **y**. - (Same, but with one variance *per group* instead of one common.) - (Same, but it predicts the *signed rank* of **y**.)	
Multiple regression: lm(y ~ 1 + x_1 + x_2 +…)	P: One-way ANOVA N: Kruskal-Wallis	aov(y ~ group) kruskal.test(y ~ group)	lm(y ~ 1 + G_2 + G_3 +…+ G_N)[A] lm(rank(y) ~ 1 + G_2 + G_3 +…+ G_N)[A]	✓ for N >11	An intercept for **group 1** (plus a difference if group ≠ 1) predicts **y**. - (Same, but it predicts the *rank* of **y**.)	
	P: One-way ANCOVA	aov(y ~ group + x)	lm(y ~ 1 + G_2 + G_3 +…+ G_N + x)[A]	✓	- (Same, but plus a slope on **x**.) *Note: this is discrete AND continuous. ANCOVAs are ANOVAs with a continuous x.*	
	P: Two-way ANOVA	aov(y ~ group * sex)	lm(y ~ 1 + G_2 + G_3 + … + G_N + S_2 + S_3+ … + S_K + G_2*S_2+G_3*S_3+…+G_N*S_K)	✓	Interaction term: changing **sex** changes the **y ~ group** parameters. *Note: **$G_{2 \text{ to } N}$** is an indicator (0 or 1) for each non-intercept levels of the **group** variable. Similarly for **$S_{2 \text{ to } K}$** for sex. The first line (with G_i) is main effect of group, the second (with S_i) for sex and the third is the **group × sex** interaction. For two levels (e.g. male/female), line 2 would just be "S_2" and line 3 would be S_2 multiplied with each G_i.*	[Coming]
	Counts ~ discrete x N: Chi-square test	chisq.test(groupXsex_table)	**Equivalent log-linear model** glm(y ~ 1 + G_2 + G_3 + … + G_N + S_2 + S_3+ … + S_K + G_2*S_2+G_3*S_3+…+G_N*S_K, family=…)[A]	✓	Interaction term: (Same as Two-way ANOVA.) *Note: Run glm using the following arguments:* `glm(model, family=poisson())` *As linear-model, the Chi-square test is $\log(y_i) = \log(N) + \log(\alpha_i) + \log(\beta_j) + \log(\alpha_i\beta_j)$ where α_i and β_j are proportions. See more info in the accompanying notebook.*	*Same as Two-way ANOVA*
	N: Goodness of fit	chisq.test(y)	glm(y ~ 1 + G_2 + G_3 +…+ G_N, family=…)[A]	✓	(Same as One-way ANOVA and see Chi-Square note.)	*1W-ANOVA*

List of common parametric (P) non-parametric (N) tests and equivalent linear models. The notation y ~ 1 + x is R shorthand for y = 1·b + a·x which most of us learned in school. Models in similar colors are highly similar, but really, notice how similar they *all* are across colors! For non-parametric models, the linear models are reasonable approximations for non-small sample sizes (see "Exact" column and click links to see simulations). Other less accurate approximations exist, e.g., Wilcoxon for the sign test and Goodness-of-fit for the binomial test. The signed rank function is `signed_rank = function(x) sign(x) * rank(abs(x))`. The variables G_i and S_i are "dummy coded" indicator variables (either 0 or 1) exploiting the fact that when Δx = 1 between categories the difference equals the slope. Subscripts (e.g., G_2 or y_1) indicate different columns in data. lm requires long-format data for all non-continuous models. All of this is exposed in greater detail and worked examples at https://lindeloev.github.io/tests-as-linear.

[A] See the note to the two-way ANOVA for explanation of the notation.
[B] Same model, but with one variance per group: `gls(value ~ 1 + G₂, weights = varIdent(form = ~1|group), method="ML")`.

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https://lindeloev.net

https://lindeloev.github.io/tests-as-linear/

With GLM we can analyse

many different types of data

using many predictors

(quantitative & qualitative)

Unified, coherent framework for data analysis with many extensions:

- **GLMM** (mixed models): accomodate data structure & variation (space, time, phylogeny)

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Unified, coherent framework for data analysis with many extensions:

- **GLMM** (mixed models): accomodate data structure & variation (space, time, phylogeny)
- **GAMM** (generalised additive models): non-linear relationships
- **Model-based multivariate** statistics

Unified, coherent framework for data analysis with many extensions:

- **GLMM** (mixed models): accomodate data structure & variation (space, time, phylogeny)
- **GAMM** (generalised additive models): non-linear relationships
- **Model-based multivariate** statistics
- **Bayesian** modelling

The Generalised Linear Model (GLM) is a particularly reasonable vantage point on statistical analyses, as **many tests and procedures are special cases** of the GLM. The downside of that (and any other) vantage point is that **we first have to climb it**. There are the morass of unfamiliar terminology, the scree slopes of probability and the cliffs of distributions. **The vista, however, is magnificent**. From the GLM, t-test, ANOVA and regression neatly arrange themselves into regular patterns, and we can see the paths leading towards the horizon: to time series analyses, Bayesian statistics, spatial statistics and so forth.

Dormann 2020

Introduction to linear models

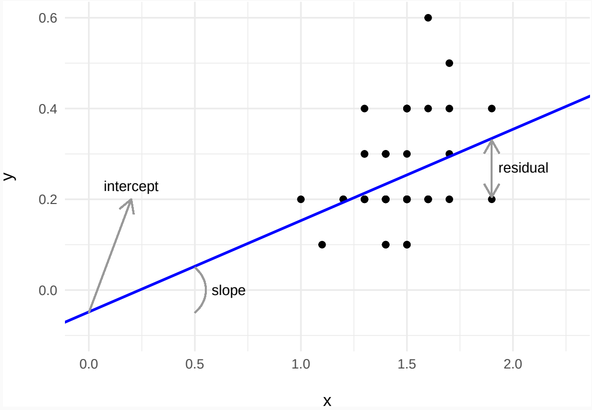
Francisco Rodríguez-Sánchez

https://frodriguezsanchez.net

Our unified regression framework (GLM)

$$y_i = a + bx_i + \varepsilon_i$$

$$\varepsilon_i \sim N\left(0, \sigma^2\right)$$



Data

y = response variable

x = predictor

Parameters

a = intercept

b = slope

σ = residual variation

ε = residuals

What’s the intercept?

Expected value of `y` when predictors (`x`) = `0`

If x = 0:

- y = a + b*0

What’s the intercept?

Expected value of `y` when predictors (`x`) = `0`

If $x = 0$:

- $y = a + b*0$
- $y = a$

What’s the slope?

How much y increases (or decreases) when x increases in 1 unit

If we have model

$y = 0.5 +$ 2*x

If x increases 1 unit, y increases **2 units**

- If x = 10 —> $y = 0.5 + 2 * 10 =$ **20.5**

What's the slope?

How much y increases (or decreases) when x increases in 1 unit

If we have model

$y = 0.5 +$ 2*x

If x increases 1 unit, y increases **2 units**

- If x = 10 —> $y = 0.5 + 2 * 10 =$ **20.5**
- If x = 11 —> $y = 0.5 + 2 * 11 =$ **22.5**

Slopes can be negative

If we have model

$y = 0.5$- 2*x

If x increases 1 unit, y decreases 2 units

- If x = 10 -> $y = 0.5 - 2 * 10 =$ -19.5

Slopes can be negative

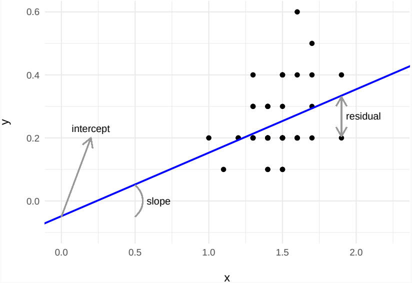
If we have model

$y =$ 0.5- 2*x

If x increases 1 unit, y decreases 2 units

- If x = 10 -> $y = 0.5 - 2 * 10 =$ -19.5
- If x = 11 -> $y = 0.5 - 2 * 11 =$ -21.5

What are residuals?



How far points fall from the regression line

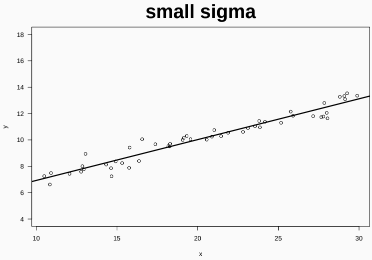
Difference between **observed values** and values **predicted** by model (regression line)

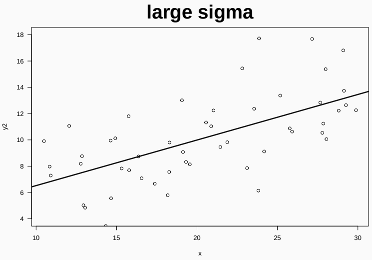
If sigma is large, residuals are larger

$$\varepsilon_i \sim N\left(0, \sigma^2\right)$$

If sigma is larger:

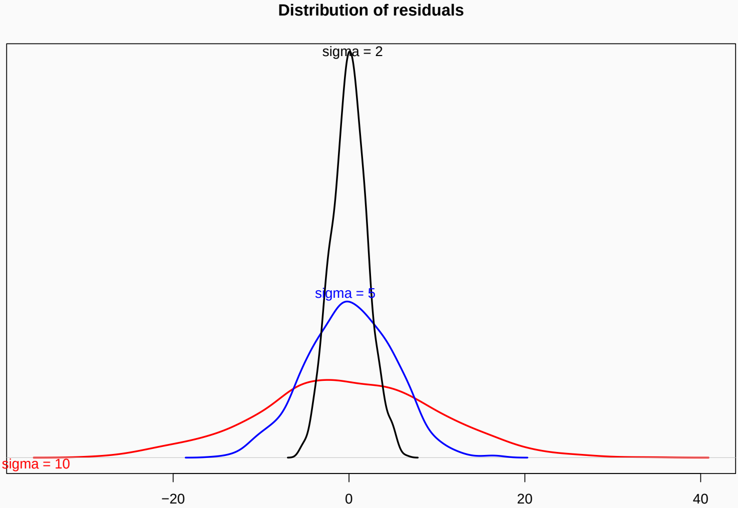
- points farther from regression line
- larger difference of observed - predicted values





Residual variation (sigma) is the Std. Dev. of residuals

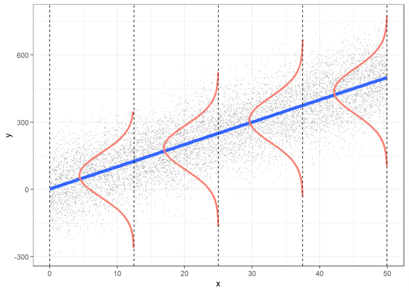
$$\varepsilon_i \sim N\left(0, \sigma^2\right)$$



In a general linear model we assume residuals are

$$\varepsilon_i \sim N\left(0, \sigma^2\right)$$

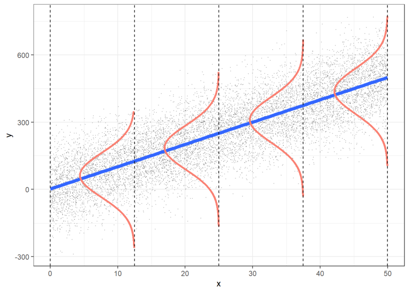
- Independent



In a general linear model we assume residuals are

$$\varepsilon_i \sim N\left(0, \sigma^2\right)$$

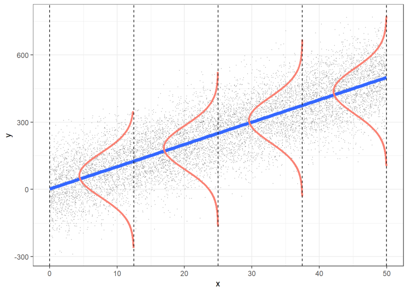
- Independent
- Normal



In a general linear model we assume residuals are

$$\varepsilon_i \sim N\left(0, \sigma^2\right)$$

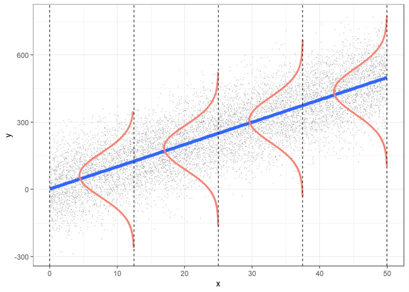
- Independent
- Normal
- Centred on 0 (no bias)



In a general linear model we assume residuals are

$$\varepsilon_i \sim N\left(0, \sigma^2\right)$$

- Independent
- Normal
- Centred on 0 (no bias)
- Homogeneous variance (*homoscedasticity*)



Different ways to write same model

$$
\begin{aligned}
y_i &= a + bx_i + \varepsilon_i \\
\varepsilon_i &\sim N\left(0, \sigma^2\right)
\end{aligned}
$$

$$
\begin{aligned}
y_i &\sim N\left(\mu_i, \sigma^2\right) \\
\mu_i &= a + bx_i \\
\varepsilon_i &\sim N\left(0, \sigma^2\right)
\end{aligned}
$$

Quiz

https://pollev.com/franciscorod726

Linear models

Francisco Rodríguez-Sánchez

https://frodriguezsanchez.net

Example dataset: forest trees

- Download this dataset (or the entire zip file)

```r
trees <- read.csv('data/trees.csv')
head(trees)
```

```
  site   dbh height    sex dead
1    4 29.68   36.1   male    0
2    5 33.29   42.3   male    0
3    2 28.03   41.9 female    0
4    5 39.86   46.5 female    0
5    1 47.94   43.9 female    0
6    1 10.82   26.2   male    0
```

Example dataset: forest trees

- Download this dataset (or the entire zip file)
- Import:

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6    1 10.82   26.2   male    0
```

- What is the relationship between DBH and height?

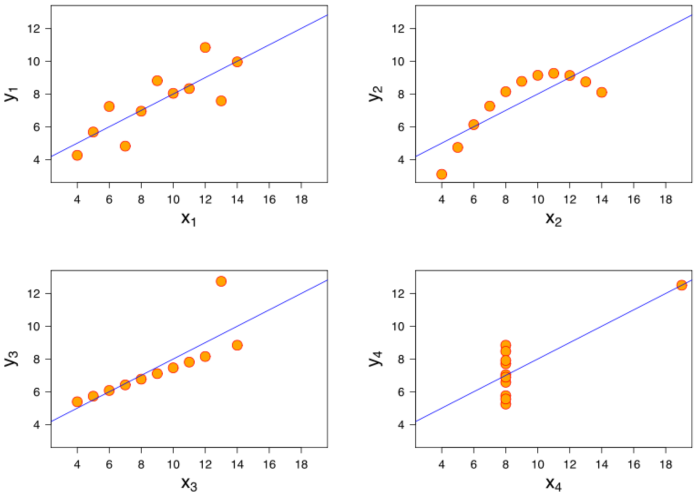
Questions

- What is the relationship between DBH and height?
- Do taller trees have bigger trunks?

- What is the relationship between DBH and height?
- Do taller trees have bigger trunks?
- Can we predict height from DBH? How well?

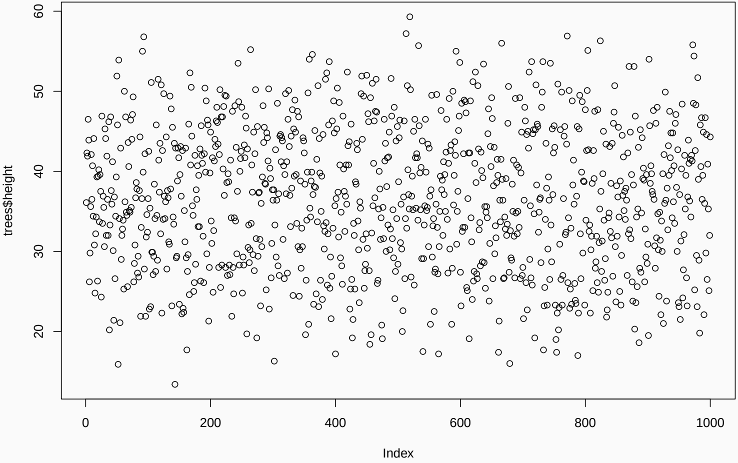
Plot your data!

Plot your data!

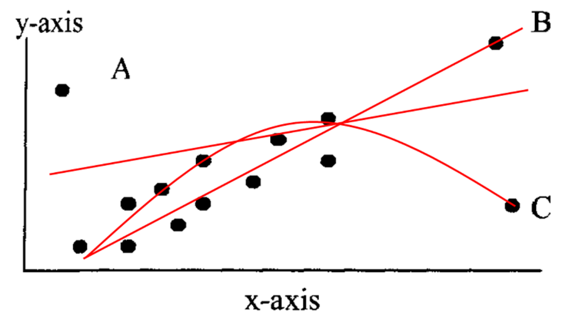


Outliers

```r
plot(trees$height)
```



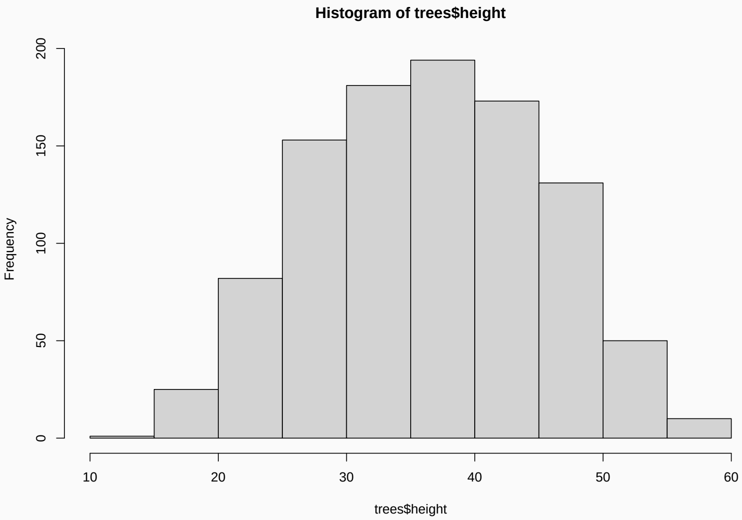
Outliers impact on regression



See http://rpsychologist.com/d3/correlation/

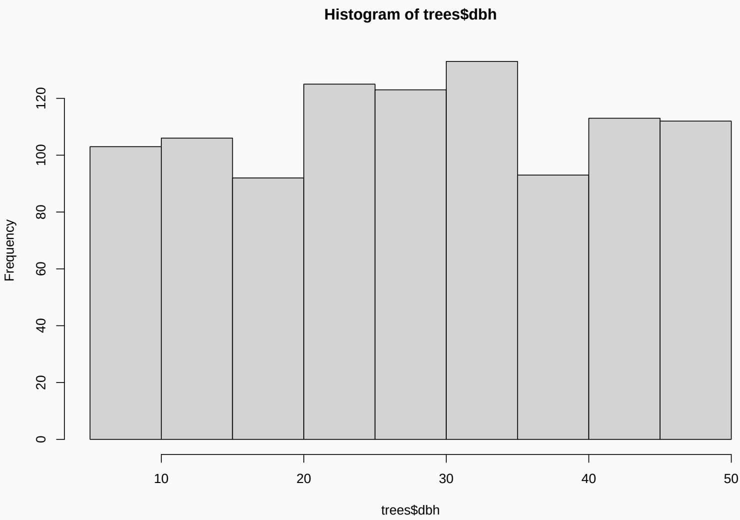
Histogram of response variable

```r
hist(trees$height)
```



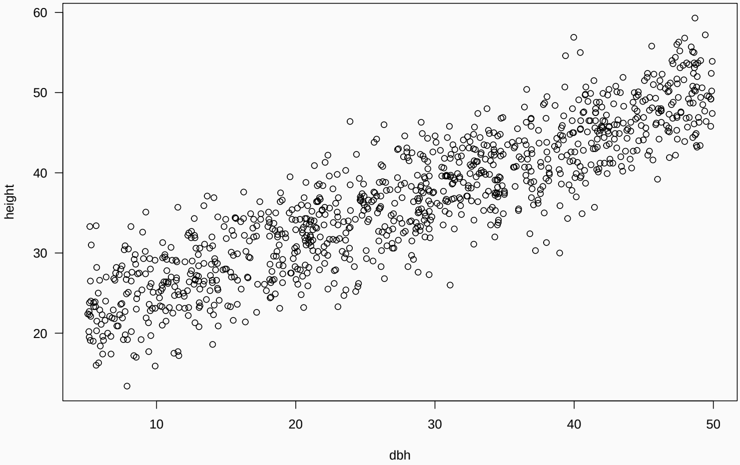
Histogram of predictor variable

```r
hist(trees$dbh)
```



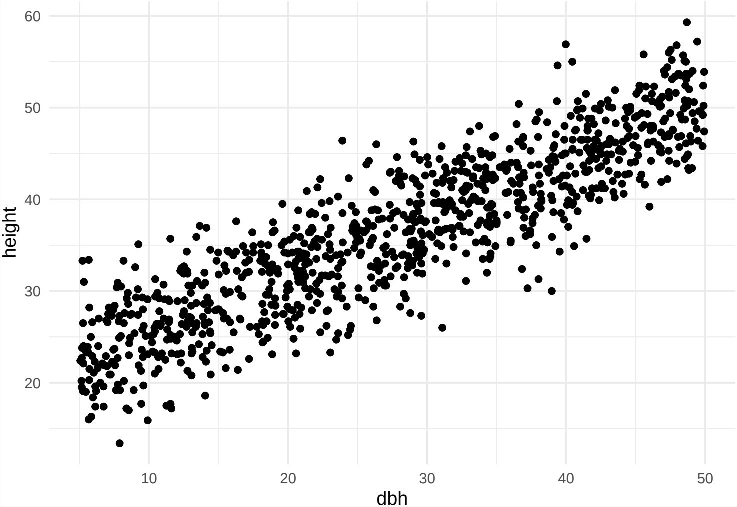
Scatterplot

```
plot(height ~ dbh, data = trees, las = 1)
```



Scatterplot

```r
ggplot(trees) +
  geom_point(aes(x = dbh, y = height))
```



Model fitting

Now fit model

Hint: `lm`

Now fit model

Hint: `lm`

```
m1 <- lm(height ~ dbh, data = trees)
```

which corresponds to

$$Height_i = a + b \cdot DBH_i + \varepsilon_i$$
$$\varepsilon_i \sim N\left(0, \sigma^2\right)$$

Package equatiomatic returns model structure

```r
library('equatiomatic')
m1 <- lm(height ~ dbh, data = trees)
equatiomatic::extract_eq(m1)
```

$$\text{height} = \alpha + \beta_1(\text{dbh}) + \epsilon \tag{1}$$

```r
equatiomatic::extract_eq(m1, use_coefs = TRUE)
```

$$\widehat{\text{height}} = 19.34 + 0.62(\text{dbh}) \tag{2}$$

To preview LaTeX:

```r
library(texPreview)
tex_preview(equatiomatic::extract_eq(m1))
```

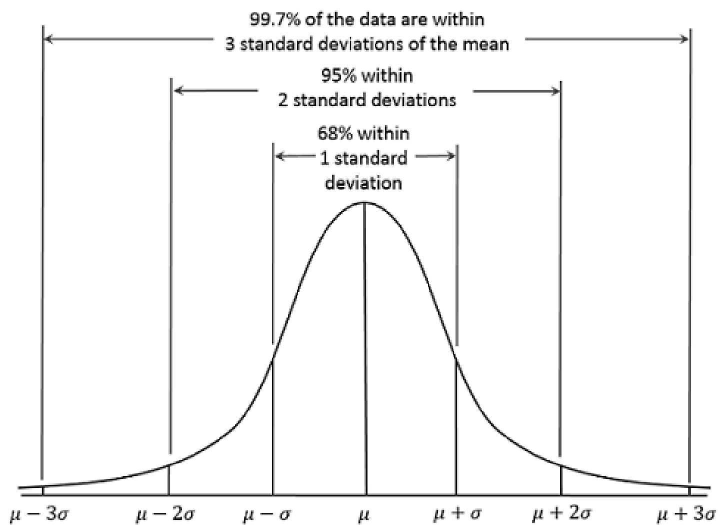
Model interpretation

What does this mean?

```r
summary(m1)
```

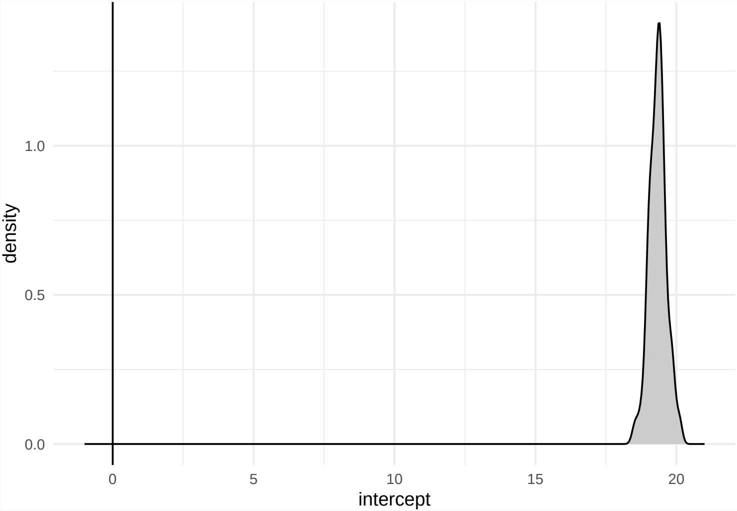
```
Call:
lm(formula = height ~ dbh, data = trees)

Residuals:
     Min       1Q   Median       3Q      Max
-13.3270  -2.8978   0.1057   2.7924  12.9511

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 19.33920    0.31064   62.26   <2e-16 ***
dbh          0.61570    0.01013   60.79   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.093 on 998 degrees of freedom
Multiple R-squared:  0.7874,    Adjusted R-squared:  0.7871
F-statistic:  3695 on 1 and 998 DF,  p-value: < 2.2e-16
```

Remember that in a Normal distribution



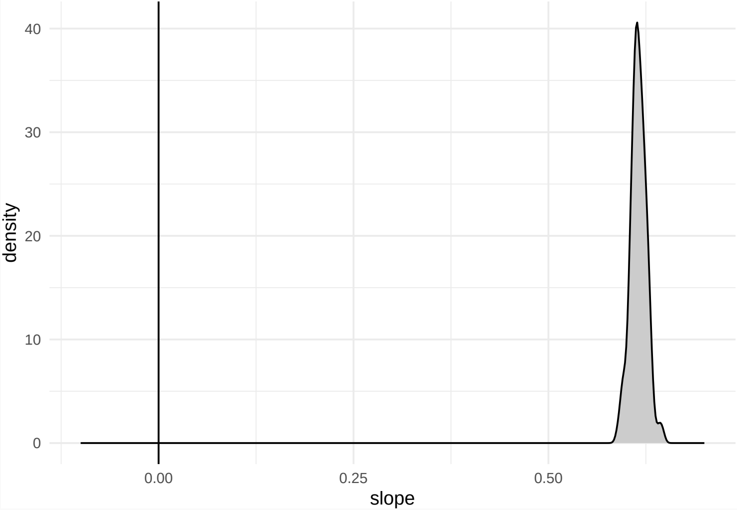
Estimated distribution of the intercept parameter

```
Parameter   | Coefficient |   SE |         95% CI | t(998) |      p
------------------------------------------------------------------
(Intercept) |       19.34 | 0.31 | [18.73, 19.95] |  62.26 | < .001
```

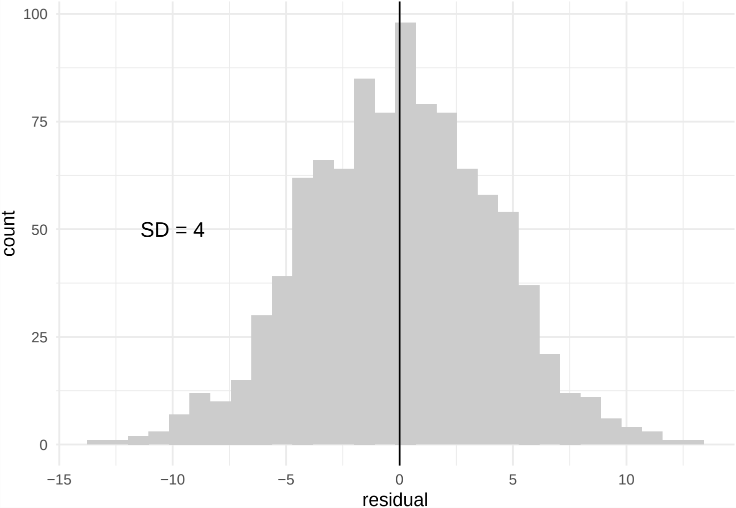


Estimated distribution of the slope parameter

```
Parameter | Coefficient |   SE |       95% CI | t(998) |      p
-----------------------------------------------------------------
dbh       |        0.62 | 0.01 | [0.60, 0.64] |  60.79 | < .001
```



Distribution of residuals



DF = n - p

n = sample size

p = number of estimated parameters

Proportion of ‘explained’ variance

$$R^2 = 1 - \frac{Residual\ Variation}{Total\ Variation}$$

Accounts for model complexity

(number of parameters)

$$R^2_{adj} = 1 - \left(1 - R^2\right)\frac{n-1}{n-p-1}$$

Quiz

https://pollev.com/franciscorod726

Centering continuous predictors

Centering continuous predictors

- Helps interpretation
- Helps (complex) model convergence
- Centering usually around the mean, but can be any suitable reference point

Centering continuous predictors

```r
mean(trees$dbh)
```

```
[1] 27.88209
```

```r
trees$dbh.c <- trees$dbh - 30
```

```r
summary(trees$dbh)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
   5.06   17.69   28.62   27.88   38.97   49.92 
```

```r
summary(trees$dbh.c)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
-24.940 -12.312  -1.380  -2.118   8.965  19.920 
```

Centering continuous predictors

```r
m1.c <- lm(height ~ dbh.c, data = trees)
```

```
Call:
lm(formula = height ~ dbh.c, data = trees)

Residuals:
     Min       1Q   Median       3Q      Max 
-13.3270  -2.8978   0.1057   2.7924  12.9511 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 37.81030    0.13119  288.22   <2e-16 ***
dbh.c        0.61570    0.01013   60.79   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.093 on 998 degrees of freedom
Multiple R-squared:  0.7874,	Adjusted R-squared:  0.7871 
F-statistic:  3695 on 1 and 998 DF,  p-value: < 2.2e-16
```

Extracting model info

Retrieving model coefficients

```r
coef(m1)
```

```
(Intercept)         dbh
 19.3391968   0.6157036
```

Confidence intervals for parameters

```r
confint(m1)
```

```
                2.5 %    97.5 %
(Intercept) 18.7296053 19.948788
dbh          0.5958282  0.635579
```

Tidy up model coefficients with broom

```r
library('broom')
tidy(m1)
```

```
# A tibble: 2 x 5
  term        estimate std.error statistic p.value
  <chr>          <dbl>     <dbl>     <dbl>   <dbl>
1 (Intercept)   19.3      0.311       62.3       0
2 dbh            0.616    0.0101      60.8       0
```

```r
glance(m1)
```

```
# A tibble: 1 x 12
  r.squared adj.r.squared sigma statistic p.value    df logLik   AIC   BIC
      <dbl>         <dbl> <dbl>     <dbl>   <dbl> <dbl>  <dbl> <dbl> <dbl>
1     0.787         0.787  4.09     3695.       0     1 -2827. 5660. 5675.
# i 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

https://broom.tidymodels.org/

Retrieving model parameters with `parameters` package

```r
library('easystats')   # parameters package
parameters(m1)
```

```
Parameter   | Coefficient |   SE |         95% CI | t(998) |      p
-------------------------------------------------------------------
(Intercept) |       19.34 | 0.31 | [18.73, 19.95] |  62.26 | < .001
dbh         |        0.62 | 0.01 | [ 0.60,  0.64] |  60.79 | < .001
```

https://easystats.github.io/parameters/

Communicating results

Avoid dichotomania of statistical significance?

MENU nature Subs

EDITORIAL · 20 MARCH 2019

It's time to talk about ditching statistical significance

- 'Never conclude there is **'no difference'** or 'no association' just because **$p >$ 0.05 or CI includes zero**'

Avoid dichotomania of statistical significance?

MENU nature Subs

EDITORIAL · 20 MARCH 2019

It's time to talk about ditching statistical significance

- 'Never conclude there is **'no difference'** or 'no association' just because **$p > 0.05$ or CI includes zero**'
- Estimate and communicate **effect sizes and their uncertainty**

Avoid dichotomania of statistical significance?

MENU nature Subs

EDITORIAL · 20 MARCH 2019

It's time to talk about ditching statistical significance

- 'Never conclude there is **'no difference'** or 'no association' just because **p > 0.05 or CI includes zero**'
- Estimate and communicate **effect sizes and their uncertainty**
- https://doi.org/10.1038/d41586-019-00857-9

Communicating results

- We found a **significant relationship** between DBH and Height **($p<0.05$)**.

Communicating results

- We found a **significant relationship** between DBH and Height **($p<0.05$)**.
- We found a {*significant*} **positive** relationship between DBH and Height {*($p<0.05$)*} **(b = 0.61, SE = 0.01)**.

- We found a **significant relationship** between DBH and Height **($p<0.05$)**.
- We found a {*significant*} **positive** relationship between DBH and Height {*($p<0.05$)*} **($b = 0.61$, $SE = 0.01$)**.
- (add p-value if you wish)

Models that describe themselves

```r
library('report')
report(m1)
```

We fitted a linear model (estimated using OLS) to predict height with dbh (formula: height ~ dbh). The model explains a statistically significant and substantial proportion of variance (R2 = 0.79, F(1, 998) = 3695.40, p < .001, adj. R2 = 0.79). The model's intercept, corresponding to dbh = 0, is at 19.34 (95% CI [18.73, 19.95], t(998) = 62.26, p < .001). Within this model:

- The effect of dbh is statistically significant and positive (beta = 0.62, 95% CI [0.60, 0.64], t(998) = 60.79, p < .001; Std. beta = 0.89, 95% CI [0.86, 0.92])

Standardized parameters were obtained by fitting the model on a standardized version of the dataset. 95% Confidence Intervals (CIs) and p-values were computed using a Wald t-distribution approximation.

https://easystats.github.io/report/

Generating table with model results: gtsummary

```
library('gtsummary')
tbl_regression(m1, intercept = TRUE)
```

Characteristic	Beta	95% CI	p-value
(Intercept)	19	19, 20	<0.001
dbh	0.62	0.60, 0.64	<0.001

Abbreviation: CI = Confidence Interval

https://www.danieldsjoberg.com/gtsummary

Generating table with model results: `modelsummary`

```r
library('modelsummary')
modelsummary(m1, output = 'markdown')  # Word, PDF, PowerPoint, png...
```

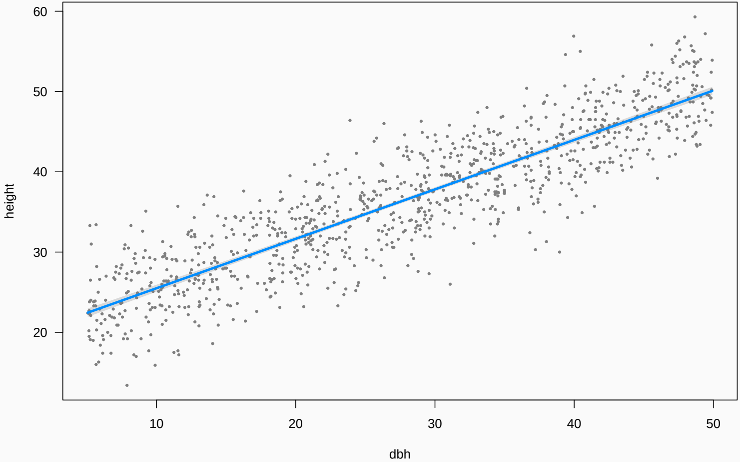
	(1)
(Intercept)	19.339
	(0.311)
dbh	0.616
	(0.010)
Num.Obs.	1000
R2	0.787
R2 Adj.	0.787
AIC	5660.3
BIC	5675.0
Log.Lik.	- 2827.125
F	3695.395
RMSE	4.09

https://modelsummary.com/

Visualising fitted model

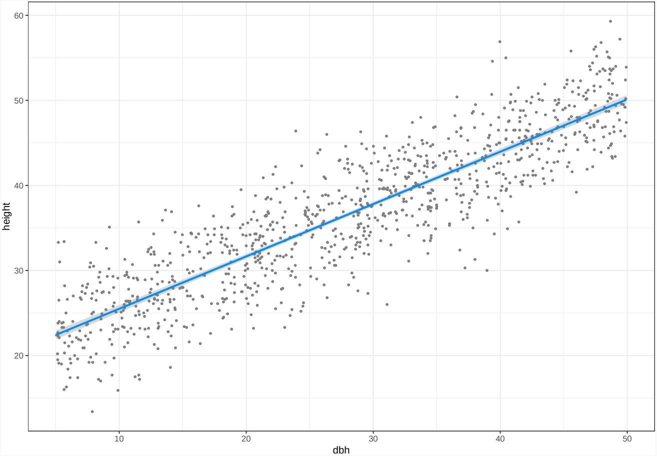
Plot model: `visreg`

```r
library('visreg')
visreg(m1)
```



visreg can use ggplot2 too

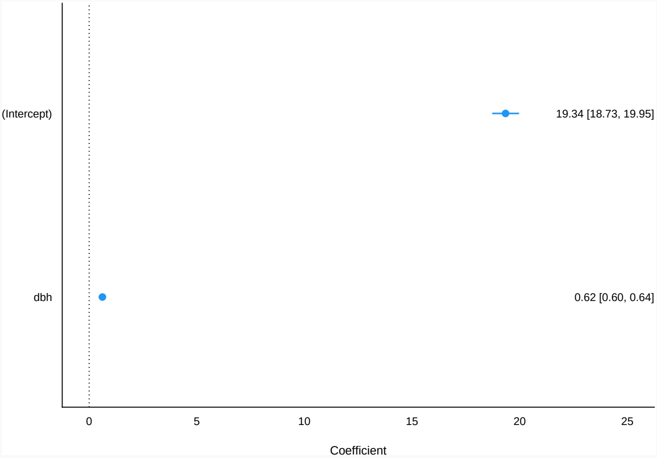
```
visreg(m1, gg = TRUE) + theme_bw()
```



https://pbreheny.github.io/visreg

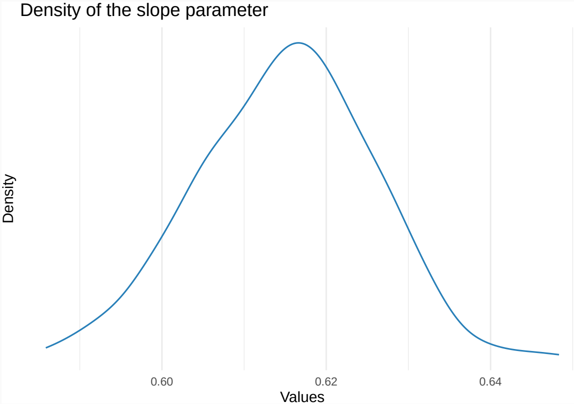
Plot model parameters with easystats (see package)

```
library('easystats')
plot(parameters(m1), show_intercept = TRUE, show_labels = TRUE)
```



Plot parameters' estimated distribution

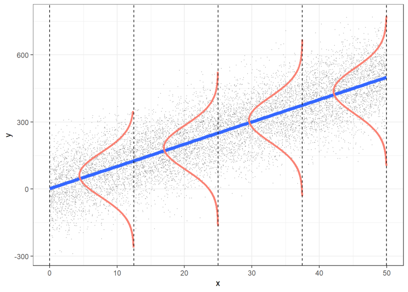
```
plot(simulate_parameters(m1)) +
  labs(title = 'Density of the slope parameter')
```



Model checking

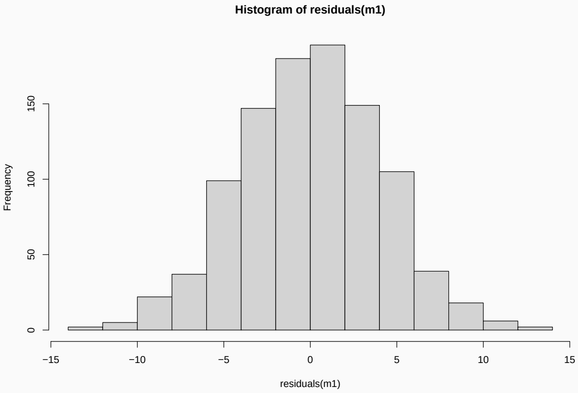
Linear model assumptions

- **Linearity** (transformations, GAM...)
- **Residuals**:
  - Independent
  - Constant variance
  - Normal
  - Mean close to zero



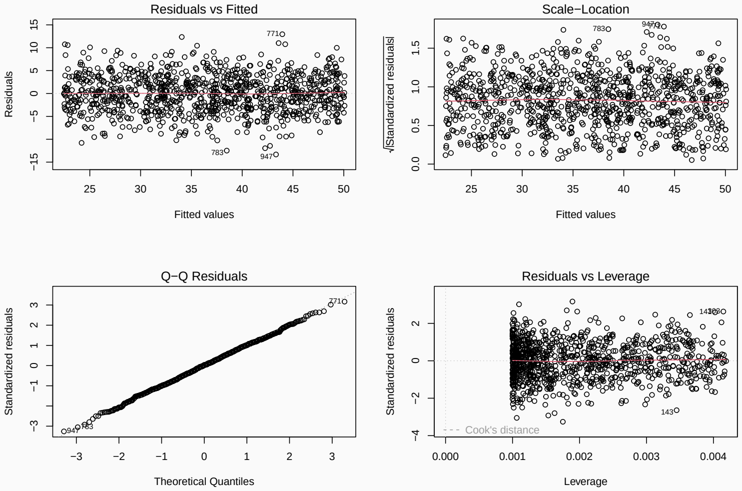
Are residuals normal?

```r
hist(residuals(m1))
```



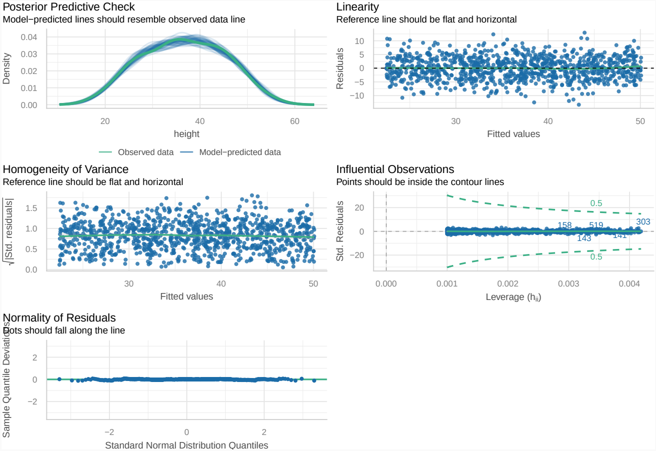
SD = 4.09

Model checking: `plot(model)`



Model checking with `performance` (easystats)

```r
library('easystats')
check_model(m1)
```



https://easystats.github.io/performance/articles/check_model.html

Using model for prediction

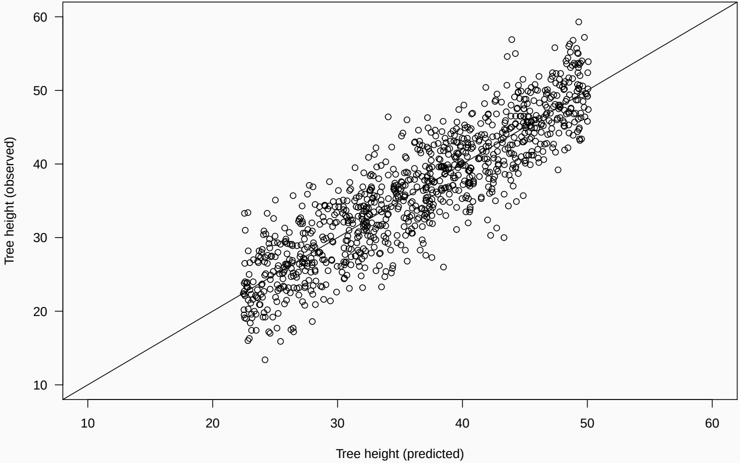
How good is the model in predicting tree height?

`fitted` gives expected value for each observation

```
trees$height.pred <- fitted(m1)
trees$resid <- residuals(m1)
head(trees)
```

```
  site   dbh height    sex dead  dbh.c height.pred      resid
1    4 29.68   36.1   male    0  -0.32    37.61328 -1.5132797
2    5 33.29   42.3   male    0   3.29    39.83597  2.4640303
3    2 28.03   41.9 female    0  -1.97    36.59737  5.3026313
4    5 39.86   46.5 female    0   9.86    43.88114  2.6188577
5    1 47.94   43.9 female    0  17.94    48.85603 -4.9560274
6    1 10.82   26.2   male    0 -19.18    26.00111  0.1988903
```

Calibration plot: Observed vs Predicted values



Making predictions for new data

Q: Expected tree height if DBH = 39 cm?

```r
new.dbh <- data.frame(dbh = c(39))
predict(m1, new.dbh, se.fit = TRUE)
```

```
$fit
       1
43.35164

$se.fit
[1] 0.1715514

$df
[1] 998

$residual.scale
[1] 4.092629
```

Confidence vs Prediction Intervals

Q: Expected tree height if DBH = 39 cm?

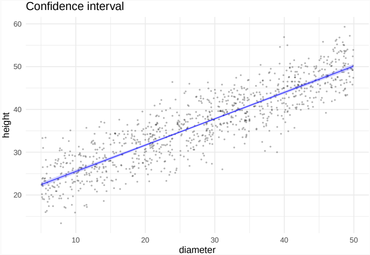
```r
predict(m1, new.dbh, interval = 'confidence')
```

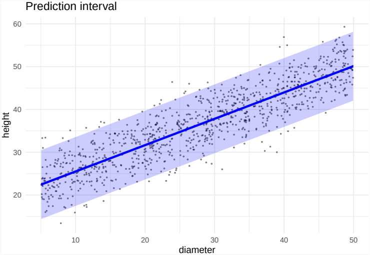
```
       fit      lwr      upr
1 43.35164 43.01499 43.68828
```

```r
predict(m1, new.dbh, interval = 'prediction')
```

```
       fit      lwr      upr
1 43.35164 35.31344 51.38983
```

Confidence vs Prediction Intervals





Workflow

- Visualise data

- Visualise data
- Understand fitted model (`summary`)

Workflow

- Visualise data
- Understand fitted model (`summary`)
- Visualise model (`visreg`...)

Workflow

- Visualise data
- Understand fitted model (`summary`)
- Visualise model (`visreg`...)
- Check model (`plot`, `check_model`, calibration plot...)

- Visualise data
- Understand fitted model (`summary`)
- Visualise model (`visreg`...)
- Check model (`plot`, `check_model`, calibration plot...)
- Predict (`predict`, `estimate_expectation`, `estimate_prediction`)

Categorical predictors (factors)

Example dataset: forest trees

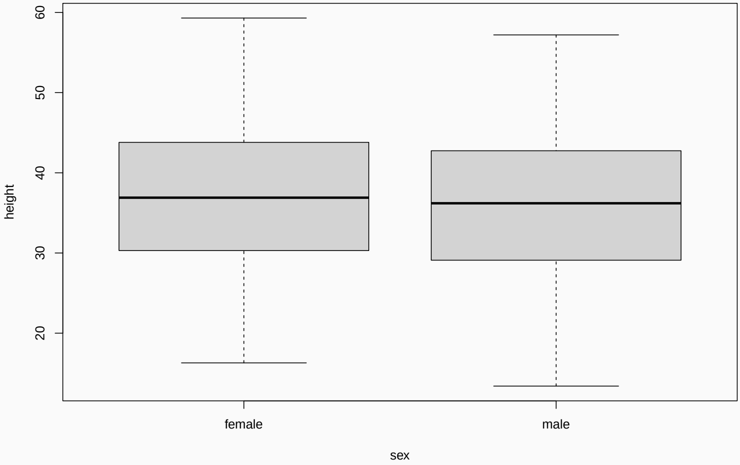
```r
trees <- read.csv('data/trees.csv')
head(trees)
```

```
  site   dbh height    sex dead
1    4 29.68   36.1   male    0
2    5 33.29   42.3   male    0
3    2 28.03   41.9 female    0
4    5 39.86   46.5 female    0
5    1 47.94   43.9 female    0
6    1 10.82   26.2   male    0
```

Does tree height vary with sex?

Q: Does tree height vary with sex?

```r
boxplot(height ~ sex, data = trees)
```



Model height ~ sex

```r
m2 <- lm(height ~ sex, data = trees)
```

```
Call:
lm(formula = height ~ sex, data = trees)

Residuals:
     Min       1Q   Median       3Q      Max 
-22.6881  -6.7881  -0.0097   6.7261  22.3687 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  36.9312     0.3981  92.778   <2e-16 ***
sexmale      -0.8432     0.5607  -1.504    0.133    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.865 on 998 degrees of freedom
Multiple R-squared:  0.002261,	Adjusted R-squared:  0.001261 
F-statistic: 2.261 on 1 and 998 DF,  p-value: 0.133
```

Linear model with categorical predictors

```r
m2 <- lm(height ~ sex, data = trees)
```

corresponds to

$$Height_i = a + b_{male} + \varepsilon_i$$
$$\varepsilon_i \sim N\left(0, \sigma^2\right)$$

Model height ~ sex

```r
m2 <- lm(height ~ sex, data = trees)
```

```
Call:
lm(formula = height ~ sex, data = trees)

Residuals:
     Min       1Q   Median       3Q      Max 
-22.6881  -6.7881  -0.0097   6.7261  22.3687 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  36.9312     0.3981  92.778   <2e-16 ***
sexmale      -0.8432     0.5607  -1.504    0.133    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.865 on 998 degrees of freedom
Multiple R-squared:  0.002261,  Adjusted R-squared:  0.001261 
F-statistic: 2.261 on 1 and 998 DF,  p-value: 0.133
```

Quiz

https://pollev.com/franciscorod726

Let's read the model report...

```r
report(m2)
```

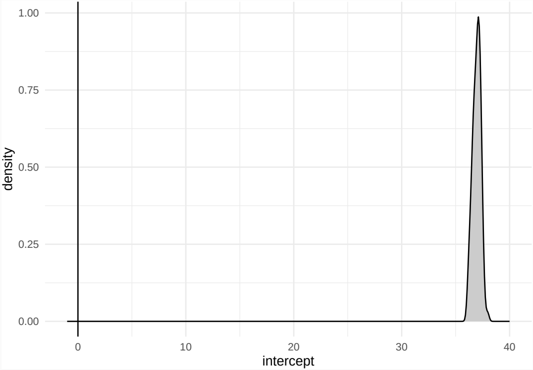
We fitted a linear model (estimated using OLS) to predict height with sex (formula: height ~ sex). The model explains a statistically not significant and very weak proportion of variance (R2 = 2.26e-03, F(1, 998) = 2.26, p = 0.133, adj. R2 = 1.26e-03). The model's intercept, corresponding to sex = female, is at 36.93 (95% CI [36.15, 37.71], t(998) = 92.78, p < .001). Within this model:

- The effect of sex [male] is statistically non-significant and negative (beta = -0.84, 95% CI [-1.94, 0.26], t(998) = -1.50, p = 0.133; Std. beta = -0.10, 95% CI [-0.22, 0.03])

Standardized parameters were obtained by fitting the model on a standardized version of the dataset. 95% Confidence Intervals (CIs) and p-values were computed using a Wald t-distribution approximation.

Intercept = Height of females

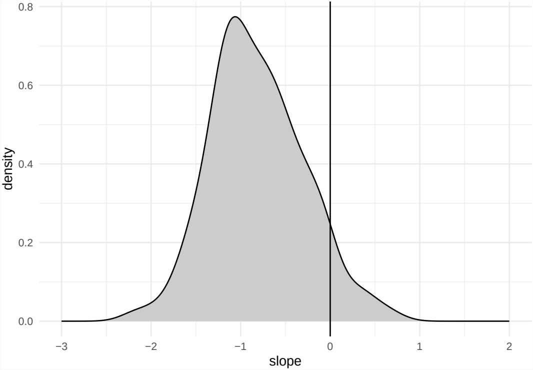
```
Parameter   | Coefficient |   SE |         95% CI | t(998) |      p
------------------------------------------------------------------
(Intercept) |       36.93 | 0.40 | [36.15, 37.71] |  92.78 | < .001
```



Estimated distribution of the *beta* parameter

beta = **height difference** of males vs females

```
Parameter  | Coefficient |   SE |        95% CI | t(998) |     p
----------------------------------------------------------------
sex [male] |       -0.84 | 0.56 | [-1.94, 0.26] |  -1.50 | 0.133
```



S-values as alternative to p-values

```r
parameters(m2)[2,]
```

```
Parameter   | Coefficient |   SE |        95% CI | t(998) |     p
-----------------------------------------------------------------
sex [male] |       -0.84 | 0.56 | [-1.94, 0.26] |  -1.50 | 0.133
```

```r
parameters(m2, s_value = TRUE)[2,]
```

```
Parameter   | Coefficient |   SE |        95% CI | t(998) |    s
----------------------------------------------------------------
sex [male] |       -0.84 | 0.56 | [-1.94, 0.26] |  -1.50 | 2.91
```

p-value = 0.133 corresponds to *surprise* of obtaining c. 3 heads in a row when tossing a coin

https://marginaleffects.com/bonus/svalues.html

```r
library('easystats')  # modelbased package
estimate_means(m2)
```

```
Estimated Marginal Means

sex    |  Mean |   SE |          95% CI | t(998)
-----------------------------------------------
female | 36.93 | 0.40 | [36.15, 37.71] |  92.78
male   | 36.09 | 0.39 | [35.31, 36.86] |  91.39

Variable predicted: height
Predictors modulated: sex
```

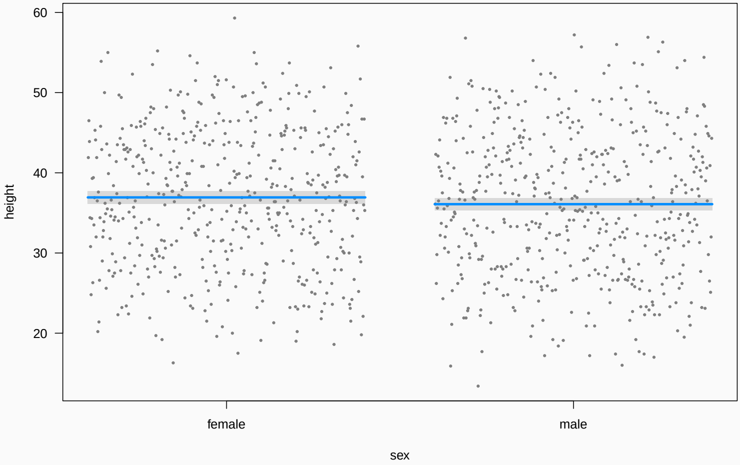
Analysing differences among factor levels

```r
estimate_contrasts(m2)
```

```
Marginal Contrasts Analysis

Level1 | Level2 | Difference |   SE |         95% CI | t(998) |     p
----------------------------------------------------------------------
male   | female |      -0.84 | 0.56 | [-1.94, 0.26] |  -1.50 | 0.133

Variable predicted: height
Predictors contrasted: sex
p-values are uncorrected.
```

Visualising the fitted model

```r
visreg(m2)
```

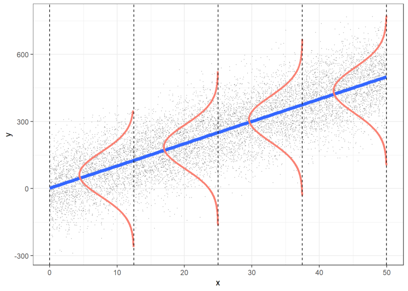


Model checking

In a general linear model we assume residuals are

$$\varepsilon_i \sim N\left(0, \sigma^2\right)$$

- Independent

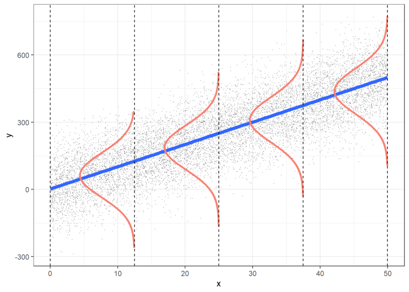


Roback & Legler 2021

In a general linear model we assume residuals are

$$\varepsilon_i \sim N\left(0, \sigma^2\right)$$

- Independent
- Normal

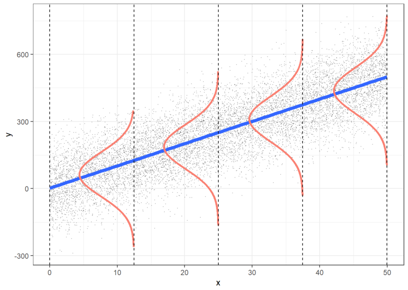


Roback & Legler 2021

In a general linear model we assume residuals are

$$\varepsilon_i \sim N\left(0, \sigma^2\right)$$

- Independent
- Normal
- Centred on 0 (no bias)

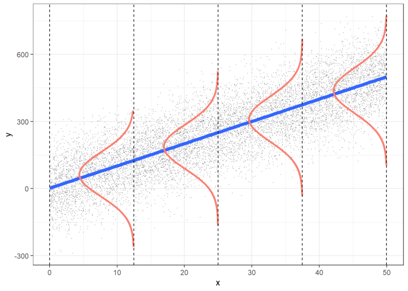


Roback & Legler 2021

In a general linear model we assume residuals are

$$\varepsilon_i \sim N\left(0, \sigma^2\right)$$

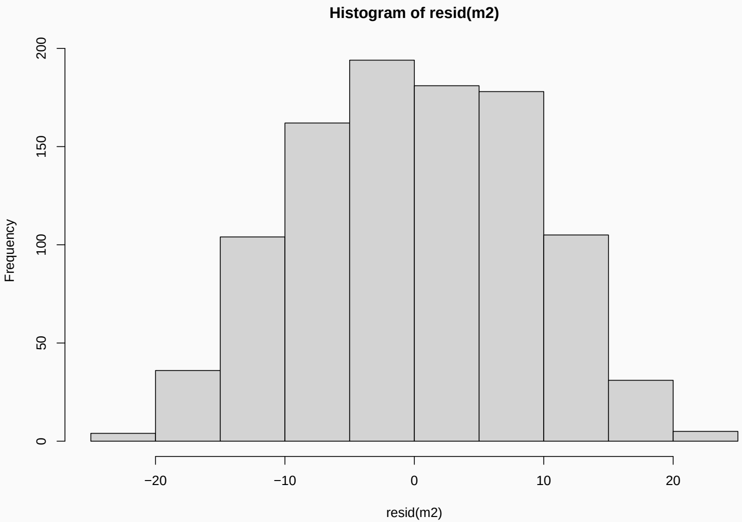
- Independent
- Normal
- Centred on 0 (no bias)
- Homogeneous variance (*homoscedasticity*)



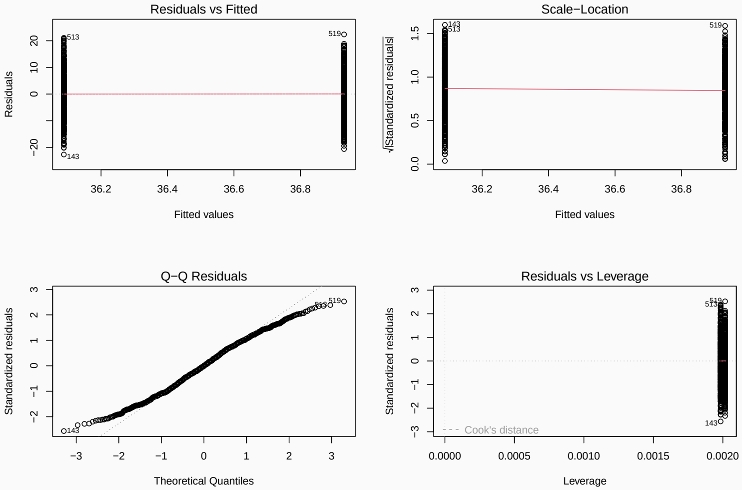
Roback & Legler 2021

Model checking: residuals

```r
hist(resid(m2))
```

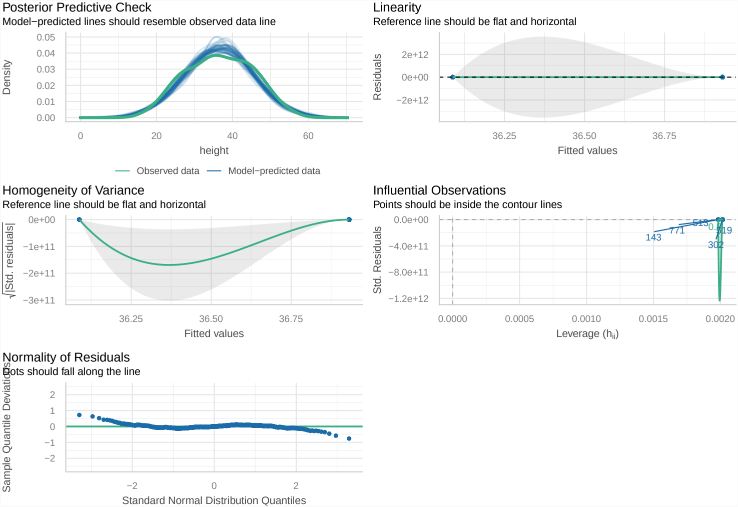


Model checking: residuals



Model checking

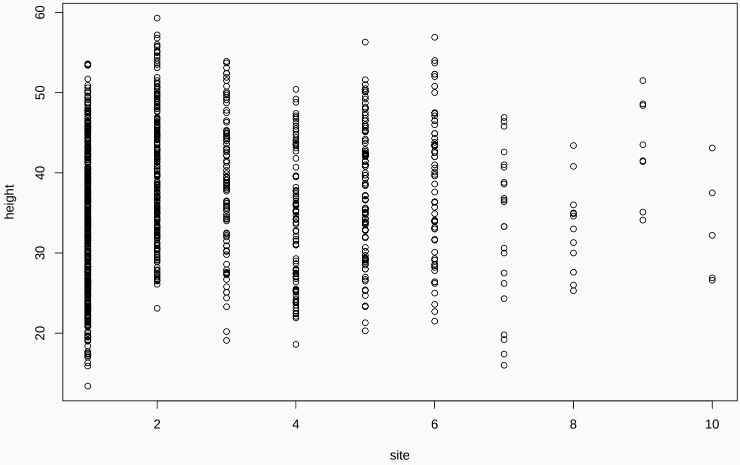
```r
library('easystats')
check_model(m2)
```



Q: Does height differ among field sites?

Plot data first

```
plot(height ~ site, data = trees)
```



Model Height ~ site

All right here?

```
m3 <- lm(height ~ site, data = trees)
```

```
Call:
lm(formula = height ~ site, data = trees)

Residuals:
     Min       1Q   Median       3Q      Max
-22.4498  -6.7049   0.0709   6.7537  23.0640

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  35.4636     0.4730  74.975  < 2e-16 ***
site          0.3862     0.1413   2.733  0.00639 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.842 on 998 degrees of freedom
Multiple R-squared:  0.007429,	Adjusted R-squared:  0.006435
F-statistic:  7.47 on 1 and 998 DF,  p-value: 0.006385
```

Let's check model structure with `equatiomatic`

```r
extract_eq(m3)
```

$$\text{height} = \alpha + \beta_{1}(\text{site}) + \epsilon \tag{3}$$

site is a factor!

```r
trees$site <- as.factor(trees$site)
```

```r
trees <- trees |>
  dplyr::mutate(site = as.factor(site))
```

Let's check model structure with `equatiomatic`

```
m3 <- lm(height ~ site, data = trees)

extract_eq(m3)
```

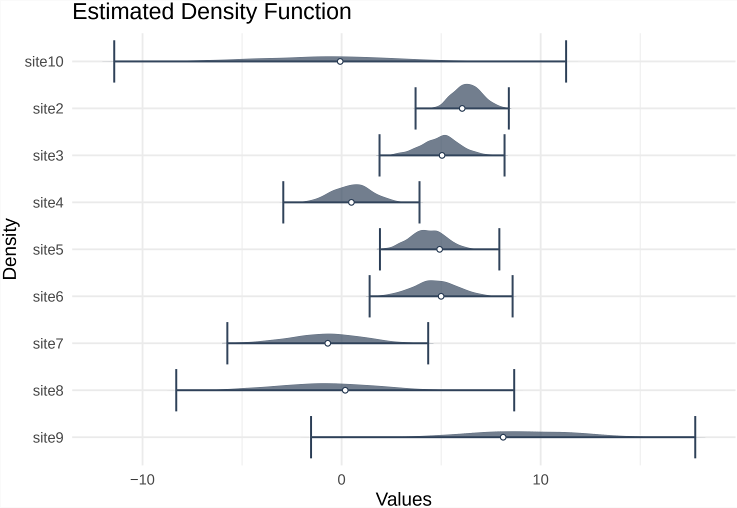
$$\text{height} = \alpha + \beta_{1}(\text{site}_{2}) + \beta_{2}(\text{site}_{3}) + \beta_{3}(\text{site}_{4}) + \beta_{4}(\text{site}_{5}) + \beta_{5}(\text{site}_{6}) + \beta_{6}(\text{site}_{7}) + \beta_{7}(\text{site}_{8}) + \beta_{8}(\text{site} \tag{4}$$

Model Height ~ site

```
Call:
lm(formula = height ~ site, data = trees)

Residuals:
     Min       1Q   Median       3Q      Max 
-20.4416  -6.9004   0.0379   6.3051  19.7584 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  33.8416     0.4266  79.329  < 2e-16 ***
site2         6.3411     0.7126   8.899  < 2e-16 ***
site3         4.9991     0.9828   5.086 4.36e-07 ***
site4         0.5329     0.9872   0.540  0.58949    
site5         4.3723     0.9425   4.639 3.97e-06 ***
site6         4.7601     1.1709   4.065 5.18e-05 ***
site7        -0.7416     1.8506  -0.401  0.68871    
site8        -0.6832     2.4753  -0.276  0.78258    
site9         9.1709     3.0165   3.040  0.00243 ** 
site10       -0.5816     3.8013  -0.153  0.87843    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.446 on 990 degrees of freedom
Multiple R-squared:  0.1016,	Adjusted R-squared:  0.09344 
F-statistic: 12.44 on 9 and 990 DF,  p-value: < 2.2e-16
```

Estimated parameter distributions

```
plot(simulate_parameters(m3), stack = FALSE)
```



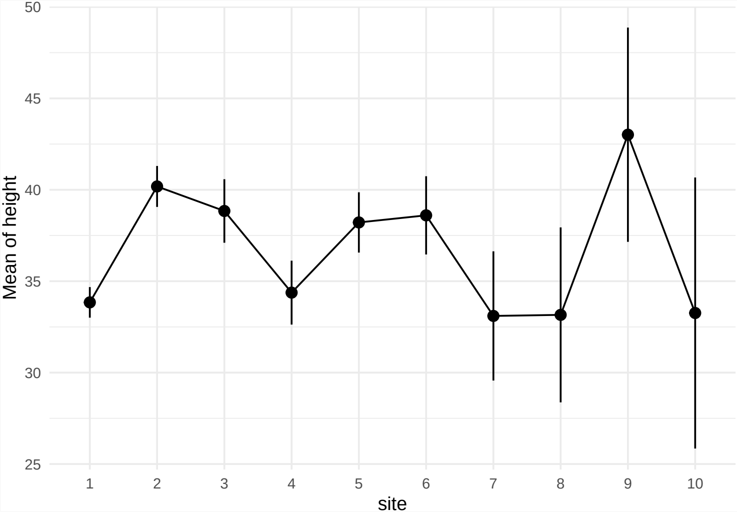
Estimated tree heights for each site

```r
estimate_means(m3)
```

```
Estimated Marginal Means

site |  Mean |   SE |         95% CI | t(990)
---------------------------------------------
1    | 33.84 | 0.43 | [33.00, 34.68] |  79.33
2    | 40.18 | 0.57 | [39.06, 41.30] |  70.40
3    | 38.84 | 0.89 | [37.10, 40.58] |  43.87
4    | 34.37 | 0.89 | [32.63, 36.12] |  38.61
5    | 38.21 | 0.84 | [36.56, 39.86] |  45.47
6    | 38.60 | 1.09 | [36.46, 40.74] |  35.40
7    | 33.10 | 1.80 | [29.57, 36.63] |  18.38
8    | 33.16 | 2.44 | [28.37, 37.94] |  13.60
9    | 43.01 | 2.99 | [37.15, 48.87] |  14.40
10   | 33.26 | 3.78 | [25.85, 40.67] |   8.81
```

Plot estimated tree heights for each site

```r
plot(estimate_means(m3))
```



Analysing differences among factor levels

For finer control see **emmeans** package

```r
estimate_contrasts(m3)
```

```
Marginal Contrasts Analysis

Level1 | Level2 | Difference |   SE |          95% CI | t(990) |      p
------------------------------------------------------------------------
2      | 1      |       6.34 | 0.71 | [  4.94,  7.74] |   8.90 | < .001
3      | 1      |       5.00 | 0.98 | [  3.07,  6.93] |   5.09 | < .001
4      | 1      |       0.53 | 0.99 | [ -1.40,  2.47] |   0.54 |  0.589
5      | 1      |       4.37 | 0.94 | [  2.52,  6.22] |   4.64 | < .001
6      | 1      |       4.76 | 1.17 | [  2.46,  7.06] |   4.07 | < .001
7      | 1      |      -0.74 | 1.85 | [ -4.37,  2.89] |  -0.40 |  0.689
8      | 1      |      -0.68 | 2.48 | [ -5.54,  4.17] |  -0.28 |  0.783
9      | 1      |       9.17 | 3.02 | [  3.25, 15.09] |   3.04 |  0.002
10     | 1      |      -0.58 | 3.80 | [ -8.04,  6.88] |  -0.15 |  0.878
3      | 2      |      -1.34 | 1.05 | [ -3.41,  0.73] |  -1.27 |  0.203
4      | 2      |      -5.81 | 1.06 | [ -7.88, -3.73] |  -5.49 | < .001
5      | 2      |      -1.97 | 1.02 | [ -3.96,  0.02] |  -1.94 |  0.053
6      | 2      |      -1.58 | 1.23 | [ -4.00,  0.83] |  -1.28 |  0.199
7      | 2      |      -7.08 | 1.89 | [-10.79, -3.38] |  -3.75 | < .001
8      | 2      |      -7.02 | 2.50 | [-11.94, -2.11] |  -2.81 |  0.005
9      | 2      |       2.83 | 3.04 | [ -3.14,  8.80] |   0.93 |  0.352
10     | 2      |      -6.92 | 3.82 | [-14.42,  0.57] |  -1.81 |  0.070
4      | 3      |      -4.47 | 1.26 | [ -6.93, -2.00] |  -3.56 | < .001
5      | 3      |      -0.63 | 1.22 | [ -3.02,  1.77] |  -0.51 |  0.608
6      | 3      |      -0.24 | 1.40 | [ -3.00,  2.52] |  -0.17 |  0.865
7      | 3      |      -5.74 | 2.01 | [ -9.68, -1.80] |  -2.86 |  0.004
8      | 3      |      -5.68 | 2.59 | [-10.77, -0.59] |  -2.19 |  0.029
9      | 3      |       4.17 | 3.11 | [ -1.94, 10.28] |   1.34 |  0.181
10     | 3      |      -5.58 | 3.88 | [-13.19,  2.03] |  -1.44 |  0.151
5      | 4      |       3.84 | 1.22 | [  1.44,  6.24] |   3.14 |  0.002
6      | 4      |       4.23 | 1.41 | [  1.46,  6.99] |   3.00 |  0.003
7      | 4      |      -1.27 | 2.01 | [ -5.22,  2.67] |  -0.63 |  0.526
```

How different are site 2 and site 9?

```r
library('marginaleffects')
hypotheses(m3, 'site2 = site9')
```

```
  Hypothesis Estimate Std. Error      z Pr(>|z|)   S 2.5 % 97.5 %
 site2=site9    -2.83       3.04 -0.931    0.352 1.5 -8.79   3.13
```

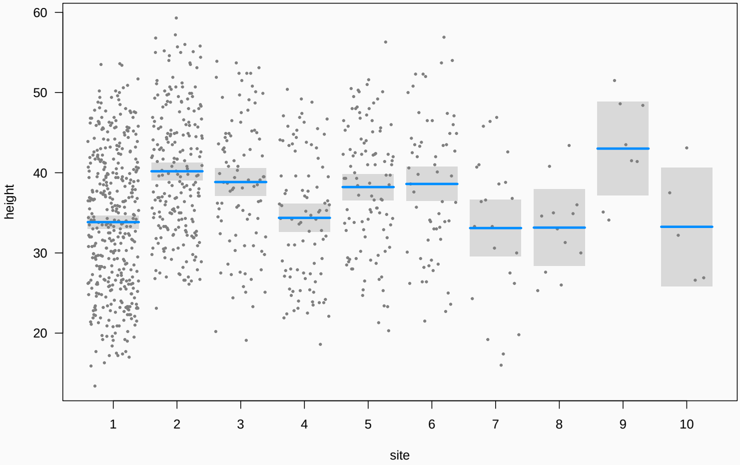
Presenting model results

```r
estimate_means(m3)
```

```
Estimated Marginal Means

site |  Mean |   SE |         95% CI | t(990)
-----------------------------------------------
1    | 33.84 | 0.43 | [33.00, 34.68] |  79.33
2    | 40.18 | 0.57 | [39.06, 41.30] |  70.40
3    | 38.84 | 0.89 | [37.10, 40.58] |  43.87
4    | 34.37 | 0.89 | [32.63, 36.12] |  38.61
5    | 38.21 | 0.84 | [36.56, 39.86] |  45.47
6    | 38.60 | 1.09 | [36.46, 40.74] |  35.40
7    | 33.10 | 1.80 | [29.57, 36.63] |  18.38
8    | 33.16 | 2.44 | [28.37, 37.94] |  13.60
9    | 43.01 | 2.99 | [37.15, 48.87] |  14.40
10   | 33.26 | 3.78 | [25.85, 40.67] |   8.81

Variable predicted: height
Predictors modulated: site
```

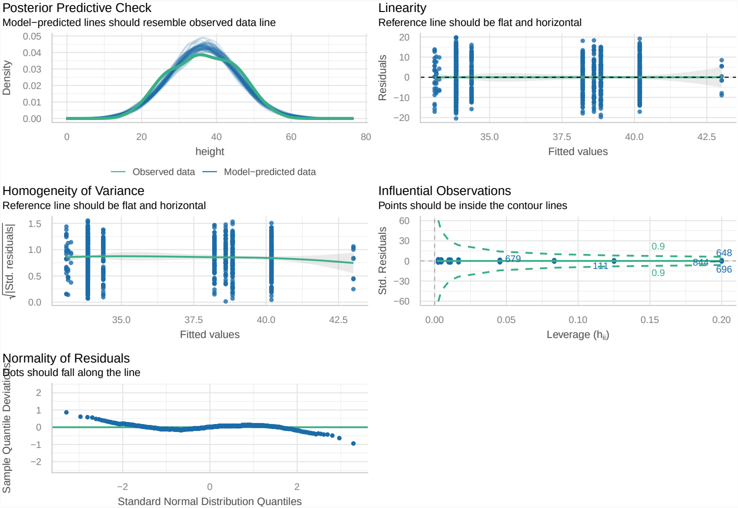
Plot (visreg)

```
visreg(m3)
```



Model checking: residuals

```r
check_model(m3)
```



Combining continuous and categorical predictors

Example dataset: forest trees

```r
trees <- read.csv("data/trees.csv")
trees$site <- as.factor(trees$site)
head(trees)
```

```
  site   dbh height    sex dead
1    4 29.68   36.1   male    0
2    5 33.29   42.3   male    0
3    2 28.03   41.9 female    0
4    5 39.86   46.5 female    0
5    1 47.94   43.9 female    0
6    1 10.82   26.2   male    0
```

Predicting tree height based on dbh and site

```r
lm(height ~ site + dbh, data = trees)
```

corresponds to

$$y_i = a + b_{site2} + c_{site3} + d_{site4} + e_{site5} + ... + k \cdot DBH_i + \varepsilon_i$$
$$\varepsilon_i \sim N\left(0, \sigma^2\right)$$

Predicting tree height based on dbh and site

```
Call:
lm(formula = height ~ site + dbh, data = trees)

Residuals:
     Min       1Q   Median       3Q      Max 
-10.1130  -1.9885   0.0582   2.0314  11.3320 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 16.699037   0.260565  64.088  < 2e-16 ***
site2        6.504303   0.256730  25.335  < 2e-16 ***
site3        4.357457   0.354181  12.303  < 2e-16 ***
site4        1.934650   0.356102   5.433 6.98e-08 ***
site5        3.637432   0.339688  10.708  < 2e-16 ***
site6        4.204511   0.421906   9.966  < 2e-16 ***
site7       -0.176193   0.666772  -0.264   0.7916    
site8       -5.312648   0.893603  -5.945 3.82e-09 ***
site9        5.437049   1.087766   4.998 6.84e-07 ***
site10       2.263338   1.369986   1.652   0.0988 .  
dbh          0.617075   0.007574  81.473  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.043 on 989 degrees of freedom
Multiple R-squared:  0.8835,    Adjusted R-squared:  0.8823
```

Let's read the model report...

```
report(m4)
```

We fitted a linear model (estimated using OLS) to predict height with site and dbh (formula: height ~ site + dbh). The model explains a statistically significant and substantial proportion of variance (R2 = 0.88, F(10, 989) = 750.04, p < .001, adj. R2 = 0.88). The model's intercept, corresponding to site = 1 and dbh = 0, is at 16.70 (95% CI [16.19, 17.21], t(989) = 64.09, p < .001). Within this model:

- The effect of site [2] is statistically significant and positive (beta = 6.50, 95% CI [6.00, 7.01], t(989) = 25.34, p < .001; Std. beta = 0.73, 95% CI [0.68, 0.79])
- The effect of site [3] is statistically significant and positive (beta = 4.36, 95% CI [3.66, 5.05], t(989) = 12.30, p < .001; Std. beta = 0.49, 95% CI [0.41, 0.57])
- The effect of site [4] is statistically significant and positive (beta = 1.93, 95% CI [1.24, 2.63], t(989) = 5.43, p < .001; Std. beta = 0.22, 95% CI [0.14, 0.30])
- The effect of site [5] is statistically significant and positive (beta = 3.64, 95% CI [2.97, 4.30], t(989) = 10.71, p < .001; Std. beta = 0.41, 95% CI [0.33, 0.49])
- The effect of site [6] is statistically significant and positive (beta = 4.20, 95% CI [3.38, 5.03], t(989) = 9.97, p < .001; Std. beta = 0.47, 95% CI [0.38, 0.57])
- The effect of site [7] is statistically non-significant and negative (beta = -0.18, 95% CI [-1.48, 1.13], t(989) = -0.26, p = 0.792; Std. beta = -0.02, 95% CI [-0.17, 0.13])
- The effect of site [8] is statistically significant and negative (beta = -5.31, 95% CI [-7.07, -3.56], t(989) = -5.95, p < .001; Std. beta = -0.60, 95% CI [-0.80, -0.40])

Presenting model results

```r
parameters(m4)
```

```
Parameter   | Coefficient |       SE |         95% CI | t(989) |      p
------------------------------------------------------------------------
(Intercept) |       16.70 |     0.26 | [16.19, 17.21] |  64.09 | < .001
site [2]    |        6.50 |     0.26 | [ 6.00,  7.01] |  25.34 | < .001
site [3]    |        4.36 |     0.35 | [ 3.66,  5.05] |  12.30 | < .001
site [4]    |        1.93 |     0.36 | [ 1.24,  2.63] |   5.43 | < .001
site [5]    |        3.64 |     0.34 | [ 2.97,  4.30] |  10.71 | < .001
site [6]    |        4.20 |     0.42 | [ 3.38,  5.03] |   9.97 | < .001
site [7]    |       -0.18 |     0.67 | [-1.48,  1.13] |  -0.26 | 0.792
site [8]    |       -5.31 |     0.89 | [-7.07, -3.56] |  -5.95 | < .001
site [9]    |        5.44 |     1.09 | [ 3.30,  7.57] |   5.00 | < .001
site [10]   |        2.26 |     1.37 | [-0.43,  4.95] |   1.65 | 0.099
dbh         |        0.62 | 7.57e-03 | [ 0.60,  0.63] |  81.47 | < .001
```

Estimated tree heights for each site

```
estimate_means(m4)
```

```
Estimated Marginal Means

site |  Mean |   SE |         95% CI | t(989)
---------------------------------------------
1    | 33.90 | 0.15 | [33.60, 34.21] | 220.59
2    | 40.41 | 0.21 | [40.01, 40.81] | 196.49
3    | 38.26 | 0.32 | [37.64, 38.89] | 119.91
4    | 35.84 | 0.32 | [35.21, 36.47] | 111.55
5    | 37.54 | 0.30 | [36.95, 38.14] | 123.94
6    | 38.11 | 0.39 | [37.34, 38.88] |  96.99
7    | 33.73 | 0.65 | [32.45, 35.00] |  51.98
8    | 28.59 | 0.88 | [26.86, 30.32] |  32.48
9    | 39.34 | 1.08 | [37.23, 41.45] |  36.53
10   | 36.17 | 1.36 | [33.50, 38.84] |  26.57
```

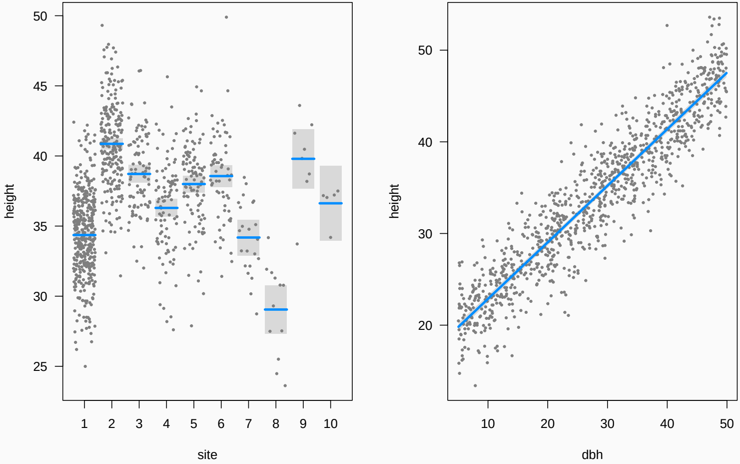
Fit model without intercept

```r
m4.noint <- lm(height ~ -1 + site + dbh, data = trees)
```

```
Call:
lm(formula = height ~ -1 + site + dbh, data = trees)

Residuals:
     Min       1Q   Median       3Q      Max 
-10.1130  -1.9885   0.0582   2.0314  11.3320 

Coefficients:
        Estimate Std. Error t value Pr(>|t|)    
site1  16.699037   0.260565   64.09   <2e-16 ***
site2  23.203340   0.292773   79.25   <2e-16 ***
site3  21.056494   0.386532   54.48   <2e-16 ***
site4  18.633687   0.374456   49.76   <2e-16 ***
site5  20.336469   0.373942   54.38   <2e-16 ***
site6  20.903548   0.448913   46.56   <2e-16 ***
site7  16.522844   0.679936   24.30   <2e-16 ***
site8  11.386389   0.918198   12.40   <2e-16 ***
site9  22.136086   1.105970   20.02   <2e-16 ***
site10 18.962375   1.372158   13.82   <2e-16 ***
dbh     0.617075   0.007574   81.47   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Plot (visreg)

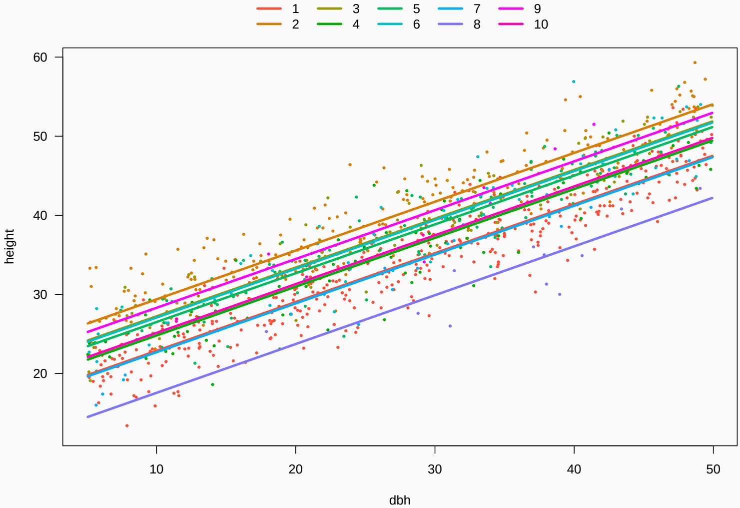
```r
visreg(m4)
```




```
null device
          1
```

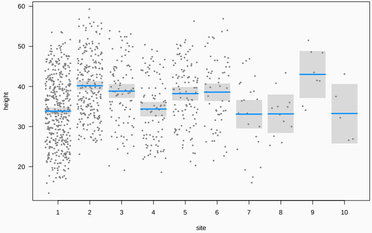
Plot (visreg)

```
visreg(m4, xvar = 'dbh', by = 'site', overlay = TRUE, band = FALSE)
```

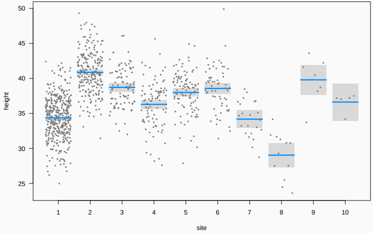


What happened to site 8?

```r
visreg(m3)
```

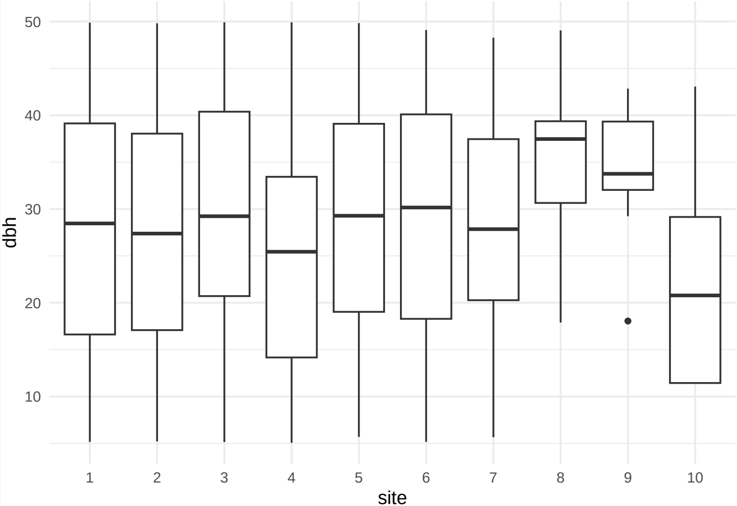



```r
visreg(m4, xvar = 'site')
```



What happened to site 8?

site 8 has the largest diameters



What happened to site 8?

DBH

```
aggregate(trees$dbh ~ trees$site, FUN = me
```

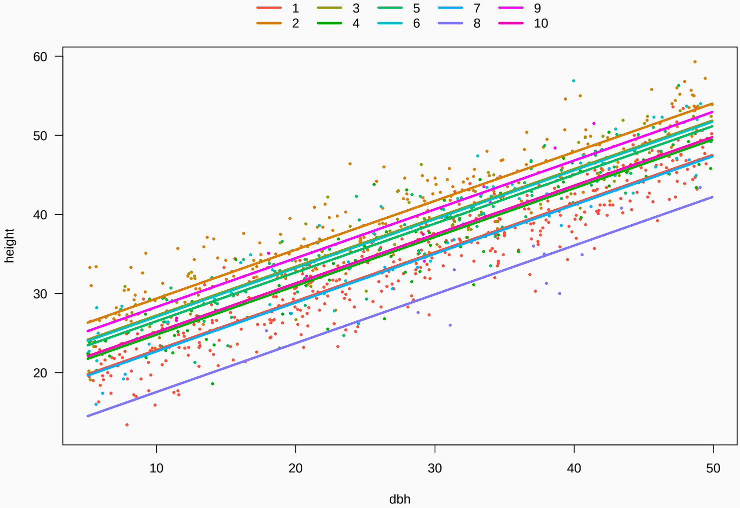
```
   trees$site trees$dbh
1           1  27.78033
2           2  27.51580
3           3  28.82011
4           4  25.50867
5           5  28.97119
6           6  28.68067
7           7  26.86409
8           8  35.28250
9           9  33.83125
10         10  23.17000
```

HEIGHT

```
aggregate(trees$height ~ trees$site, FUN =
```

```
   trees$site trees$height
1           1     33.84158
2           2     40.18265
3           3     38.84066
4           4     34.37444
5           5     38.21386
6           6     38.60167
7           7     33.10000
8           8     33.15833
9           9     43.01250
10         10     33.26000
```

We have fitted model w/ many intercepts and single slope



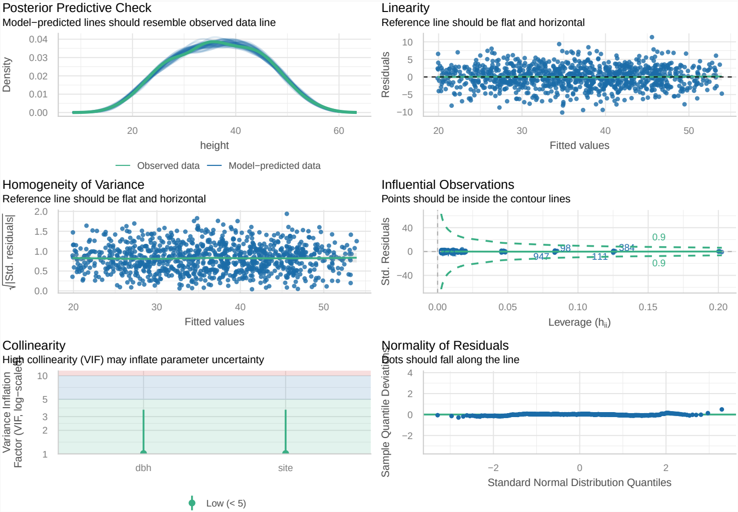
Slope is the same for all sites

```r
parameters(m4, keep = 'dbh')
```

```
Parameter | Coefficient |       SE |       95% CI | t(989) |      p
-------------------------------------------------------------------
dbh       |        0.62 | 7.57e-03 | [0.60, 0.63] |  81.47 | < .001
```

Model checking: residuals

```r
check_model(m4)
```



Predicting heights of new trees

Using model for prediction

Expected height of 10-cm diameter tree in each site?

```r
trees.10cm <- data.frame(site = as.factor(1:10),
                         dbh = 10)
trees.10cm
```

```
   site dbh
1     1  10
2     2  10
3     3  10
4     4  10
5     5  10
6     6  10
7     7  10
8     8  10
9     9  10
10   10  10
```

Using model for prediction

Confidence interval

```r
predict(m4, newdata = trees.10cm, interval = 'confidence')
```

```
        fit      lwr      upr
1  22.86979 22.46878 23.27079
2  29.37409 28.89388 29.85430
3  27.22724 26.54160 27.91289
4  24.80444 24.13410 25.47477
5  26.50722 25.84952 27.16492
6  27.07430 26.25490 27.89370
7  22.69359 21.39601 23.99117
8  17.55714 15.79282 19.32146
9  28.30683 26.16606 30.44761
10 25.13312 22.45540 27.81085
```

Using model for prediction

Prediction interval (accounting for residual variance)

```r
predict(m4, newdata = trees.10cm, interval = 'prediction')
```

```
        fit      lwr      upr
1  22.86979 16.88478 28.85480
2  29.37409 23.38325 35.36493
3  27.22724 21.21645 33.23804
4  24.80444 18.79537 30.81350
5  26.50722 20.49955 32.51489
6  27.07430 21.04678 33.10181
7  22.69359 16.58268 28.80451
8  17.55714 11.33039 23.78388
9  28.30683 21.96314 34.65053
10 25.13312 18.58868 31.67757
```

Using model for prediction

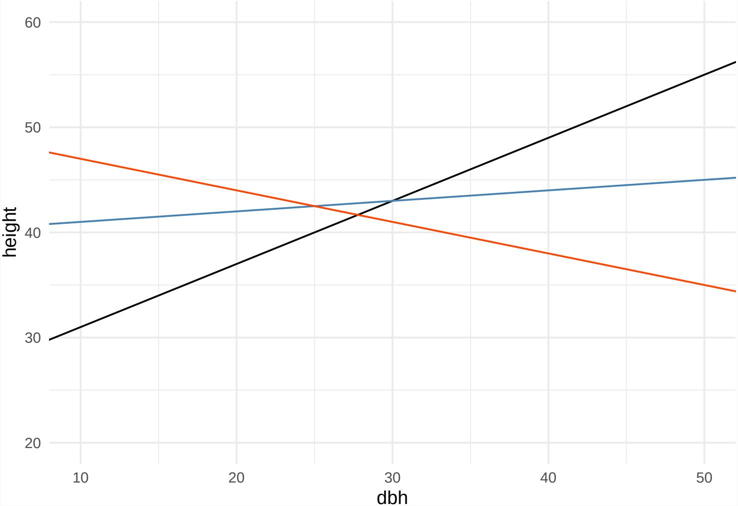
Prediction interval (99%)

```r
predict(m4, newdata = trees.10cm, interval = 'prediction',
        level = 0.99)
```

```
        fit       lwr      upr
1  22.86979 14.998587 30.74098
2  29.37409 21.495225 37.25295
3  27.22724 19.322133 35.13235
4  24.80444 16.901598 32.70727
5  26.50722 18.606216 34.40822
6  27.07430 19.147195 35.00140
7  22.69359 14.656813 30.73037
8  17.55714  9.368019 25.74626
9  28.30683 19.963913 36.64976
10 25.13312 16.526183 33.74007
```

Q: Does allometric relationship between Height and Diameter vary among sites?

Does allometric relationship between Height and Diameter vary among sites?

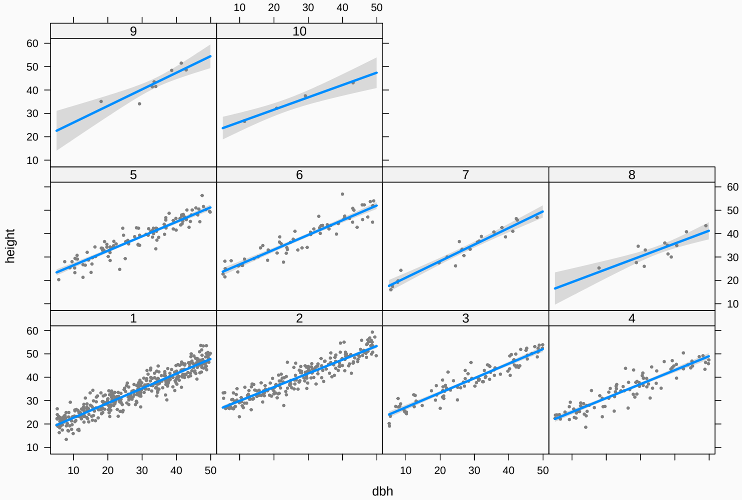


Model with interactions

```
Call:
lm(formula = height ~ site * dbh, data = trees)

Residuals:
     Min       1Q   Median       3Q      Max 
-10.1017  -1.9839   0.0645   2.0486  11.1789 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 16.359437   0.360054  45.436  < 2e-16 ***
site2        7.684781   0.609657  12.605  < 2e-16 ***
site3        4.518568   0.867008   5.212 2.28e-07 ***
site4        2.769336   0.813259   3.405 0.000688 ***
site5        3.917607   0.870983   4.498 7.68e-06 ***
site6        4.155161   1.009379   4.117 4.17e-05 ***
site7       -2.306799   1.551303  -1.487 0.137334    
site8       -2.616095   4.090671  -0.640 0.522630    
site9        2.621560   5.073794   0.517 0.605492    
site10       4.662340   2.991072   1.559 0.119378    
dbh          0.629299   0.011722  53.685  < 2e-16 ***
site2:dbh   -0.042784   0.020033  -2.136 0.032950 *  
site3:dbh   -0.006031   0.027640  -0.218 0.827312    
site4:dbh   -0.031633   0.028225  -1.121 0.262677    
site5:dbh   -0.010173   0.027887  -0.365 0.715334    
site6:dbh    0.001337   0.032109   0.042 0.966797    
site7:dbh    0.079728   0.052056   1.532 0.125951    
site8:dbh   -0.079027   0.113386  -0.697 0.485984    
site9:dbh    0.081035   0.146649   0.553 0.580679    
site10:dbh  -0.101107   0.114520  -0.883 0.377522    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.041 on 980 degrees of freedom
Multiple R-squared:  0.8847,	Adjusted R-squared:  0.8825
```

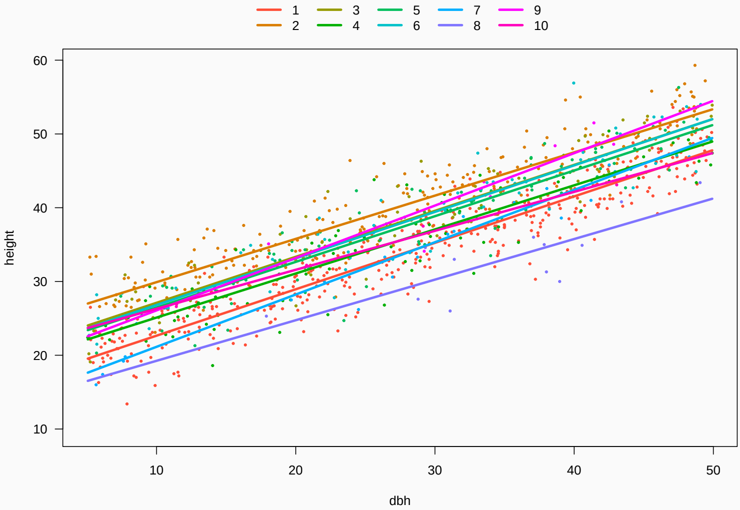
Does slope vary among sites?

```r
visreg(m5, xvar = 'dbh', by = 'site')
```



Does slope vary among sites?

```
visreg(m5, xvar = 'dbh', by = 'site', overlay = TRUE, band = FALSE)
```



Does slope vary among sites?

```r
library('marginaleffects')
hypotheses(m5, '`site9:dbh` = `site10:dbh`')
```

```
               Hypothesis Estimate Std. Error     z Pr(>|z|)   S  2.5 % 97.5 %
 `site9:dbh`=`site10:dbh`    0.182      0.185 0.983    0.326 1.6 -0.181  0.545
```